

Chapter – 7

Transportation in Steel Industry

7.1 Introduction

Steel Making is very large and integrated system. The success of steel production largely depends upon the mode of transportation and their regular supply of various feeding points. Similarly in the process of making steel lot of movements such as hot metal, slag, heat trains, finished steel, scrap etc. to be done on scheduled basis. The success of smooth production basically depends on their movements.

The finished products also get the priority in despatching them to the customer to get the continuous inflow of money, which is the main objective of any organisation.

7.2 Layout of Internal Transportation System in ISPs

Layout of traffic system network should be such that without delaying the smooth and safe operation takes place. To design an ideal layout, a careful thinking should be given keeping in mind the various parameters such as incoming raw material, their monthly and daily plan, unloading plan, shunting arrangements etc.

Similarly loading bays and tracks in shipping bays should be given careful thinking for quick loading to avoid delays in loading because loading delays results in heavy detention of wagons increase in turnaround times and consequent demurrage.

Liaison with Railways

Success of any transportation largely depends on the close co-ordination of the industry with Railways. Since Railways is the key supplier of wagons for Raw material loading such as Coal, Iron Ore, Lime stone, Dolomite, Manganese Ore etc., a very close liaison is to be maintained with Railways.

For this co-ordination meeting are taking place on quarterly/monthly basis between Railways and plants. These meetings are helpful for meeting plant requirements in time. The monthly & quarterly requirements of plant are given to Railways well in advance to fulfill industries requirement at regular intervals. The performance by plant and Railways are jointly reviewed in meeting and problems are sorted out and strategies for better movements are decided in those meetings.

Similarly for despatching of finished product and bye-product on daily basis and in monthly basis, indent for the rakes as per destinations are given by PPC and TRM to railways for placement of empty wagons for loading. Handing over the rakes to railways within the stipulated time is also being done to maintain the turn around times and reduce the demurrage.. The success of despatch largely depends on the close co-ordination of loading and handing over of the rakes in time. The rakes are normally of 59 wagons in case of BOXN type with single destination or 43 type (concurrent) with multiple destination.

This results in continuous inflow and out flow of wagons.

Road Transport

Though the steel industry largely transport their material by Rail, many times to cater the customers at door step dispatches are done by road. The dispatches by road is a requirement of the steel plants to meet the target of saleable steel.. Many

times low quantity and shorter distance makes it economical to use road transport. This system is faster also. The railways normally discourage movement of low quantity of raw material, over shorter distance.

The road transport proves many times very useful for steel plants.

Demurrage

Demurrage is charge levied by Railways for detaining Railway Wagon inside the plant area beyond specified time allowed for particular type of wagon.

Railways impose this charge for deviation from the stipulated turn round types for the place rakes. The lesser the turn round Railways earn more freight and plant pays less demurrage.

Demurrage control

To control the demurrage, the wagons should be unloaded in allotted free time and return the wagons back to Railways. The efficient handling of wagons largely depend upon quick marking of wagons, sending them to respective siding, unloading, removal of wagons by forming rake and handing them over to Railways.

Since steel plant has a very large network of tracks and siding daily inventory is to be taken and all information regarding empty and unloaded wagons to be analyzed and decision for removal and unloading of wagons should be taken with the help of information generated by centralised inventory and demurrage control unit.

The demurrage is an unproductive expenditure and also affects the rake availability, meeting are being done at different levels in these plants to avoid detention of wagons and to reduce demurrages.

Shipping Practices

The job of steel plant is not complete till we have sold our finished product to the customer. The loading and despatch largely depend on shipment department. The shipping department is having with them the Rolling plan for various mills and loading plan which is given to them by CMO/local marketing department. As per loading programme shipping department ask Traffic department for supply of wagon. Shipping department organise loading documentation etc.

Despatch Advice

After loading is over Shipping department issue Despatch Advice which indicates about Name of consignee, Type of material, quantity of material, destination, number of pieces, their approximate weight etc.

After loading is over with certificate of Railway TXR these wagons are removed from siding, weighed, forwarding note is prepared. The forwarding note is given to Railway goods clerk who after carefully checking, invoices the wagon. The rake is formed and handed over to Railways for movement to respective destination.

7.3 Role of Supervisors in Controlling Cost Related to Transportation

Normal work of any integrated steel plant is possible only when its transport system works well. Any failure in the functioning of this system will disrupt the working of the entire plant. The transport system connects one shop of the steel plant with the

other shop and is itself connected with all of them. This system serves all the units and all the production units in turn are dependent on it.

To enable the transportation system to play its role properly, it is necessary that the production units of the steel plant maintain their production parameters according to certain schedule and provide advance information about this to the transport department so that the later can also prepare movement schedules matched with each other and compact schedule is made, and if the operation is done according to this, economy in production is achieved.

It is therefore, necessary that production departments prepare their schedule in advance in consultant with Production Planning Control department and TRM so that it can also prepare its movement schedule to serve the production units as well.

The supervisor must adhere strictly to these production schedules as well as the movement schedule to make production and despatch of materials smooth and effective.

A strict discipline should be maintained in upkeep of Railway tracks/fitness of Wagons and locomotives.

7.4 Logistics of SAIL Units

1. Bhilai Steel Plant

Name of the Mines	Loading Station	Commodity	Distance via Rail (in Km)
Dalli	Kondey Siding	Iron Ore	90
Rajhara	Rajhara Siding	Iron Ore	86
Nandini (HSLD)	Hindustan Steel Plant Siding	Flux	23
Hirri	do	Flux	135
Kuteshwar	Khanabanjari	Flux	496
Rowghat (New Project)	Rowghat (Anantgarh)	Iron Ore	190 (approx..)

Table:- Distance of Captive Mines from Bhilai Steel Plant via Rail

Name of the Ports	Loading Station	Commodity	Distance via Rail (in Km)
Vishakhapatnam	Vishakhapatnam	Imported Coal	556
Gangavaram	Gangavaram	Imported Coal	561

Table:- Distance of Serving Ports from Bhilai Steel Plant via Rail

2. Durgapur Steel Plant

S.No.	Place	Commodity	Distance (km) via Rail
1.	Bolani (BYX)	Iron Ore	318
2.	Gua (Captive Mine)		309
3.	Kuteshwar (Captive Mine) (KTE)	Limestone	873
4.	Birmitrapur (BRMP)	Dolomite	354
5.	Hirri (Captive Mine) (DPH)		640
6.	Dhanbad (BCCL/CCL) Area	Domestic power/coking coal	92

Table:- Distance of Captive Mines from Durgapur Steel Plant via Rail

S.No.	Place	Commodity	Distance (km) via Rail
1.	Haldia	Imported Coking Coal and Limestone	308
2.	Dhamra		483

Table:- Distance of Serving Ports from Durgapur Steel Plant via Rail

3. Rourkela Steel Plant

Name of the Mines	Loading Station	Commodity	Distance via Rail (in Km)
Barsuan and Tilda	Barsuan	Iron ore	73
Kalta	Roxy	Iron ore	63
Kiruburu	Kiruburu	Iron ore	91
Meghataburu	Meghataburu	Iron ore	91
Biramitrapur	Biramitrapur	Limestone	35
Sonakhan	Sonakhan	Limestone	45

Table:- Distance of Captive Mines from Rourkela Steel Plant via Rail

S.No.	Name of Port	Commodity	Distance via Rail (Km)
1	Haldia	Coking Coal & Limestone	402
2	Paradip		505
3	Dhamra		527
4	Gopalpur		581

Table:- Distance of Serving Ports from Rourkela Steel Plant via Rail

4. Bokaro Steel Plant

Name of the Mines	Loading Station	Commodity	Distance via Rail (in Km)
Gua	Gua	Iron ore	264
Chirria	Manoharpur	Iron ore	240
Kiruburu	Kiruburu	Iron ore	362
Meghataburu	Meghataburu	Iron ore	362

Table:- Distance of Captive Mines from Bokaro Steel Plant via Rail

Name of the Ports	Loading Station	Commodity	Distance via Rail (in Km)
Haldia	Haldia	Coking Coal	366
Dhamra	Dhamra	Coking Coal	496
Paradip	Paradip	Coking Coal	589

Table:- Distance of Serving Ports from Bokaro Steel Plant via Rail

5. IISCO Steel Plant

S.No.	Place	Material	Distance (km) via Rail
1.	Jamadoba (JBO)	Domestic Coking Coal	76
2.	Bhelatand (PBWB)		90
3.	Bolani (BYX)	Iron Ore	285
	Gua (Captive Mine)		276
4.	Meghataburu (-do-)		376
5.	Sonu	Domestic Limestone	1944
6.	Biramitrapur (BRMP)		322
7.	Kuteshwar (Captive Mine) (KTE)		852
8.	Hirri (Captive Mine) (DPH)		617

Table:- Distance of Captive Mines from IISCO Steel Plant via Rail

S.No.	Place	Material	Distance (km) via Rail
1.	Haldia	Imported Coal	316
2.	Dhamra		451
3.	Paradip		544

Table:- Distance of Serving Ports from IISCO Steel Plant via Rail
